

CARE-SAV: A Conditioning-Aware Random-Feature Framework for Energy-Stable Simulation of Gradient Flows

Bingcheng Hu (胡秉承)

Shanghai Normal University

Joint work with Z. Li

Shanghai SIAM 2026, Shanghai, China

Sept. 13, 2026

Table of Contents

- 1 Motivation
- 2 Gradient flow and SAV
- 3 Conditioning-aware representation enhancement
- 4 The CARE-SAV method
- 5 Main results
- 6 Numerical experiments
- 7 Takeaways

Table of Contents

- 1 Motivation
- 2 Gradient flow and SAV
- 3 Conditioning-aware representation enhancement
- 4 The CARE-SAV method
- 5 Main results
- 6 Numerical experiments
- 7 Takeaways

Gradient flows and the numerical challenge

Gradient flows describe irreversible relaxation in phase-field models, interfacial dynamics and materials science:

$$\partial_t \phi = \mathcal{G} \frac{\delta E}{\delta \phi}, \quad \frac{d}{dt} E(\phi) = \left(\frac{\delta E}{\delta \phi}, \mathcal{G} \frac{\delta E}{\delta \phi} \right) \leq 0.$$

- $E(\phi)$ is a Lyapunov functional; its decay is crucial for reliable long-time simulation.
- SAV methods give efficient, linear and unconditionally energy-stable time stepping.
- Conventional fully discrete SAV schemes rely on a **prescribed** grid, finite-element, or spectral space.

Question

Can we use a flexible feature-generated spatial representation *and* retain the discrete energy-dissipation law?

CARE-SAV at a glance

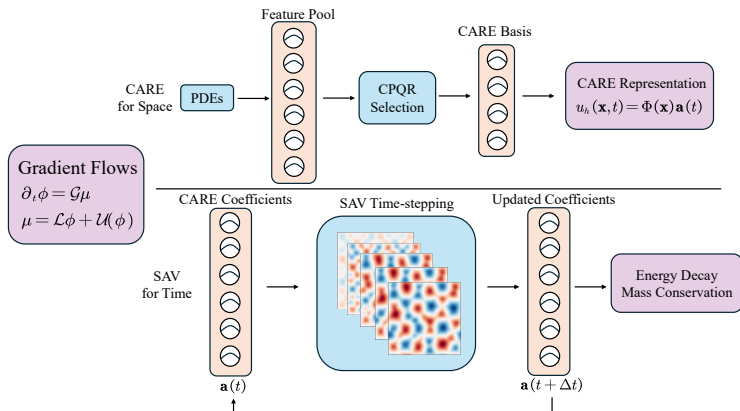


Fig.: Feature generation, conditioning-aware reduction, CARE Galerkin projection, and SAV time integration.

Table of Contents

- 1 Motivation
- 2 Gradient flow and SAV
- 3 Conditioning-aware representation enhancement
- 4 The CARE-SAV method
- 5 Main results
- 6 Numerical experiments
- 7 Takeaways

Gradient-flow formulation

Consider the energy $E(\phi) = \frac{1}{2}(\phi, \mathcal{L}\phi) + E_1(\phi)$, where $\mathcal{L}$ is self-adjoint and nonnegative. The associated system is

$$\partial_t \phi = \mathcal{G}\mu, \quad \mu = \mathcal{L}\phi + \mathcal{U}(\phi), \quad \mathcal{U}(\phi) = \frac{\delta E_1}{\delta \phi},$$

with $(\mathcal{G}v, v) \leq 0$.

- $\mathcal{G} = -I$: nonconserved flows (e.g., Allen–Cahn).
- $\mathcal{G} = \Delta$: conserved flows (e.g., Cahn–Hilliard).
- Periodic or homogeneous Neumann boundary conditions eliminate boundary terms.

Scalar auxiliary variable (SAV) reformulation

Assume $E_1(\phi) + C_0 > 0$ and introduce

$$r(t) = \sqrt{E_1(\phi) + C_0}, \quad b(\phi) = \frac{\mathcal{U}(\phi)}{\sqrt{E_1(\phi) + C_0}}.$$

Then the nonlinear gradient flow is equivalently written as

$$\partial_t \phi = \mathcal{G}\mu, \quad \mu = \mathcal{L}\phi + rb(\phi), \quad \frac{dr}{dt} = \frac{1}{2}(b(\phi), \partial_t \phi).$$

The modified energy satisfies

$$\frac{d}{dt} \left[\frac{1}{2}(\phi, \mathcal{L}\phi) + r^2 - C_0 \right] = (\mathcal{G}\mu, \mu) \leq 0.$$

Key advantage

The nonlinear energy is represented by one scalar variable, enabling a linear energy-stable update.

Table of Contents

- 1 Motivation
- 2 Gradient flow and SAV
- 3 Conditioning-aware representation enhancement
- 4 The CARE-SAV method
- 5 Main results
- 6 Numerical experiments
- 7 Takeaways

Random-feature trial space

Generate an overcomplete set of M Gaussian features:

$$\varphi_j(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_j\|^2}{2\sigma_j^2}\right), \quad V_M^{\text{RF}} = \text{span}\{\varphi_1, \dots, \varphi_M\}.$$

For periodic problems, use periodized Gaussian features. At quadrature nodes $\{\mathbf{x}_q, w_q\}_{q=1}^N$, form $\Phi_{qj} = \varphi_j(\mathbf{x}_q)$ and $\hat{\Phi} = W^{1/2}\Phi$.

- Features provide a flexible, mesh-free representation.
- Random candidates may be strongly correlated or numerically redundant.
- Direct use of V_M^{RF} can therefore be severely ill-conditioned.

CARE: a stable orthonormal Galerkin space

Use column-pivoted, rank-revealing QR:

$$\widehat{\Phi} P = QR, \quad K = \max \left\{ k : \frac{|R_{kk}|}{|R_{11}|} \geq \tau_{QR} \right\}.$$

With $T = P_K R_K^{-1}$, define $\Psi = \Phi T$. Then

$$W^{1/2} \Psi = Q_K, \quad \Psi^T W \Psi = I_K, \quad V_K^{\text{CARE}} = \text{span}\{\psi_1, \dots, \psi_K\}.$$

CARE principle

Retain numerically independent directions, discard redundant features, and obtain an identity discrete mass matrix.

- $K \leq M$ is chosen adaptively by the QR tolerance.
- The CARE basis is built once before time integration and then fixed.

CARE Galerkin projection

Seek $\phi_K = \boldsymbol{\psi}^T \mathbf{c}$ and $\mu_K = \boldsymbol{\psi}^T \mathbf{d}$ in V_K^{CARE} . With the quadrature inner product $(\cdot, \cdot)_Q$,

$$(\partial_t \phi_K, v_K)_Q = (\mathcal{G} \mu_K, v_K)_Q, \quad (\mu_K, v_K)_Q = (\mathcal{L} \phi_K, v_K)_Q + (\mathcal{U}(\phi_K), v_K)_Q.$$

The coefficient system is

$$\dot{\mathbf{c}} = \mathbf{G}_K \mathbf{d}, \quad \mathbf{d} = \mathbf{L}_K \mathbf{c} + \mathbf{U}_K(\mathbf{c}),$$

where $(\mathbf{G}_K)_{ij} = (\mathcal{G} \psi_j, \psi_i)_Q$ and $(\mathbf{L}_K)_{ij} = (\mathcal{L} \psi_j, \psi_i)_Q$.

Structural point

The orthonormal CARE basis turns the reduced mass matrix into I_K , so it is a well-conditioned Galerkin discretization.

Table of Contents

- 1 Motivation
- 2 Gradient flow and SAV
- 3 Conditioning-aware representation enhancement
- 4 The CARE-SAV method**
- 5 Main results
- 6 Numerical experiments
- 7 Takeaways

Fully discrete CARE-SAV scheme

Define $\mathbf{b}^{n+1/2} = \mathbf{b}_K((3\mathbf{c}^n - \mathbf{c}^{n-1})/2)$. The Crank–Nicolson scheme is

$$\begin{aligned}\frac{\mathbf{c}^{n+1} - \mathbf{c}^n}{\Delta t} &= \mathbf{G}_K \mathbf{d}^{n+1/2}, \\ \mathbf{d}^{n+1/2} &= \mathbf{L}_K \frac{\mathbf{c}^{n+1} + \mathbf{c}^n}{2} + \frac{r^{n+1} + r^n}{2} \mathbf{b}^{n+1/2}, \\ \frac{r^{n+1} - r^n}{\Delta t} &= \frac{1}{2} (\mathbf{b}^{n+1/2})^T \frac{\mathbf{c}^{n+1} - \mathbf{c}^n}{\Delta t}.\end{aligned}$$

- The nonlinear vector is explicit, hence the update is linear in $(\mathbf{c}^{n+1}, r^{n+1})$.
- $\tilde{E}_K^n = \frac{1}{2} (\mathbf{c}^n)^T \mathbf{L}_K \mathbf{c}^n + (r^n)^2 - C_0$.

Efficient solution procedure

Eliminating $r^{n+1/2}$ yields a rank-one correction:

$$\mathbf{A}_K \mathbf{c}^{n+1} = \mathbf{g}^n + \frac{\Delta t}{4} \alpha^{n+1/2} \mathbf{G}_K \mathbf{b}^{n+1/2}, \quad \mathbf{A}_K = \mathbf{I}_K - \frac{\Delta t}{2} \mathbf{G}_K \mathbf{L}_K.$$

Solve two systems sharing the same matrix:

$$\mathbf{A}_K \mathbf{c}_*^{n+1} = \mathbf{g}^n, \quad \mathbf{A}_K \mathbf{q}^{n+1/2} = \mathbf{G}_K \mathbf{b}^{n+1/2}.$$

Then

$$\alpha^{n+1/2} = \frac{(\mathbf{b}^{n+1/2})^T (\mathbf{c}_*^{n+1} - \mathbf{c}^n)}{1 - \frac{\Delta t}{4} (\mathbf{b}^{n+1/2})^T \mathbf{q}^{n+1/2}}, \quad \mathbf{c}^{n+1} = \mathbf{c}_*^{n+1} + \frac{\Delta t}{4} \alpha^{n+1/2} \mathbf{q}^{n+1/2}.$$

Efficiency

For fixed Δt , factorize $\mathbf{A}_K$ once; every step needs two re-used linear solves plus vector operations.

Table of Contents

- 1 Motivation
- 2 Gradient flow and SAV
- 3 Conditioning-aware representation enhancement
- 4 The CARE-SAV method
- 5 Main results**
- 6 Numerical experiments
- 7 Takeaways

Theoretical guarantees

Properties of CARE-SAV

Under the stated assumptions, the fully discrete method has:

- **unconditional unique solvability** for every $\Delta t > 0$;
- **unconditional energy stability**,

$$\tilde{E}_K^{n+1} - \tilde{E}_K^n = \Delta t (\mathbf{G}_K \mathbf{d}^{n+1/2}, \mathbf{d}^{n+1/2}) \leq 0;$$

- **mass conservation** for Cahn–Hilliard dynamics;
- **fully discrete error estimates** combining CARE-space approximation and second-order temporal accuracy.

The CPQR construction simultaneously gives a compact representation and controls the conditioning of the discrete basis.

Unique Solvability

Assumption

We assume that the reduced operators satisfy

$$\mathbf{L}_K = \mathbf{L}_K^T \succeq 0, \quad \mathbf{z}^T \mathbf{G}_K \mathbf{z} \leq 0 \quad \forall \mathbf{z} \in \mathbb{R}^K. \quad (1)$$

Unconditional Unique Solvability

Let $\mathbf{b}^{n+1/2} = \mathbf{b}_K(\bar{\mathbf{c}}^{n+1/2})$ be fixed by explicit extrapolation. Under Assumption 1, the fully discrete CARE-SAV scheme has a unique solution

$$(\mathbf{c}^{n+1}, \mathbf{d}^{n+1/2}, r^{n+1}) \in \mathbb{R}^K \times \mathbb{R}^K \times \mathbb{R}$$

for every $\Delta t > 0$.

Energy Stability

Define the discrete modified SAV energy by

$$\tilde{E}_K^n = \frac{1}{2}(\mathbf{c}^n)^T \mathbf{L}_K \mathbf{c}^n + (r^n)^2 - C_0.$$

Unconditional Energy Stability

Every solution of the fully discrete CARE-SAV scheme satisfies

$$\tilde{E}_K^{n+1} - \tilde{E}_K^n = \Delta t (\mathbf{d}^{n+1/2})^T \mathbf{G}_K \mathbf{d}^{n+1/2} \leq 0.$$

Hence, CARE-SAV preserves the energy-dissipation law for **any** time step $\Delta t > 0$.

Mass Conservation and Conditioning

Conditional Mass Conservation

Let $m_i = (1, \psi_i)_Q$ and $\mathcal{M}_Q^n = \mathbf{m}^T \mathbf{c}^n$. If $\mathbf{m}^T \mathbf{G}_K = 0$, then

$$\mathcal{M}_Q^{n+1} = \mathcal{M}_Q^n \quad \text{for every time step.}$$

Thus, CARE-SAV conserves the discrete mass for Cahn–Hilliard dynamics under the structure-preserving assembly condition.

Conditioning of the CARE Basis

The CARE basis satisfies $\Psi^T W \Psi = I_K$; therefore, its discrete mass matrix has condition number $\kappa_2 = 1$ in exact arithmetic.

Fully Discrete Error Estimate

Convergence Result

For sufficiently smooth exact solutions and sufficiently small Δt ,

$$\begin{aligned} \max_{t_n \leq T} (\|\phi(t_n) - \phi_K^n\|_{L^2} + |r(t_n) - r^n|) \\ \leq C_{T,K} [\Delta t^2 + \mathfrak{A}_{M,N,K}(T) + \mathfrak{Q}_{M,N,K}(T)] . \end{aligned}$$

- The method is **second-order accurate in time** for a fixed CARE space.
- The spatial error is governed by the approximation quality of the CARE basis and quadrature consistency.

Table of Contents

- 1 Motivation
- 2 Gradient flow and SAV
- 3 Conditioning-aware representation enhancement
- 4 The CARE-SAV method
- 5 Main results
- 6 Numerical experiments**
- 7 Takeaways

Benchmark suite and accuracy

Five representative gradient-flow models are tested: Allen–Cahn (AC), Cahn–Hilliard (CH), phase-field crystal (PFC), molecular beam epitaxy (MBE), and grain growth (GG).

Model	Challenging parameter	Relative L^2 error	CPU time (s)
AC	$\varepsilon = 10^{-5}$	4.5126×10^{-5}	17.270
CH	$\varepsilon = 0.1$	3.7604×10^{-7}	85.892
PFC	$r = -0.5$	5.8187×10^{-4}	133.85
MBE	manufactured solution	6.0786×10^{-8}	106.26

- CARE-SAV accurately resolves both nonconserved and conserved flows.
- Physical and modified SAV energies are nearly indistinguishable and decay monotonically for unforced cases.

Allen–Cahn: solution accuracy

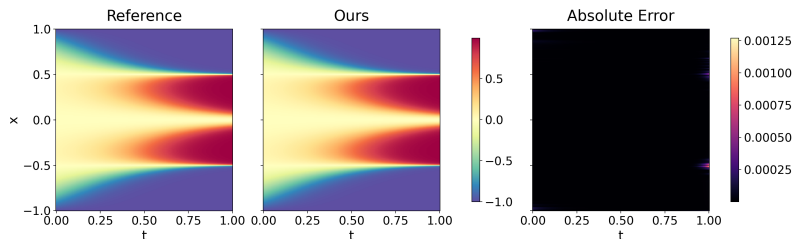


Fig.: Reference solution, CARE-SAV approximation, and pointwise absolute error for $\varepsilon = 10^{-5}$.

Allen–Cahn: energy dissipation

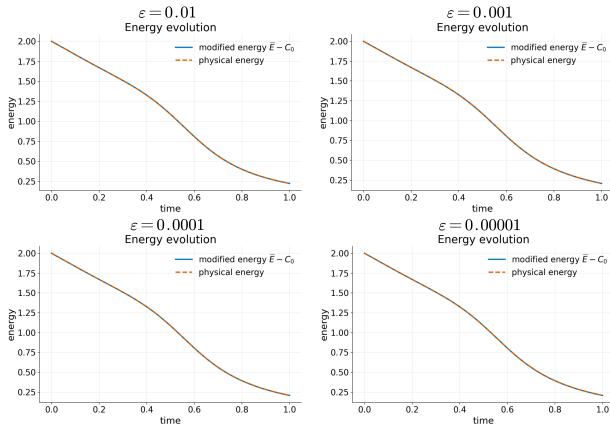


Fig.: Physical and modified SAV energies for the Allen–Cahn equation. Both decay monotonically throughout the simulation.

Cahn–Hilliard: spinodal decomposition

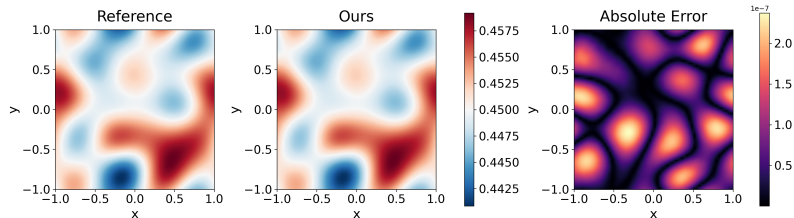


Fig.: Reference solution, CARE-SAV approximation, and pointwise absolute error for $\varepsilon = 0.1$ at $t = T$.

Phase-field crystal: resolving complex spatial structure

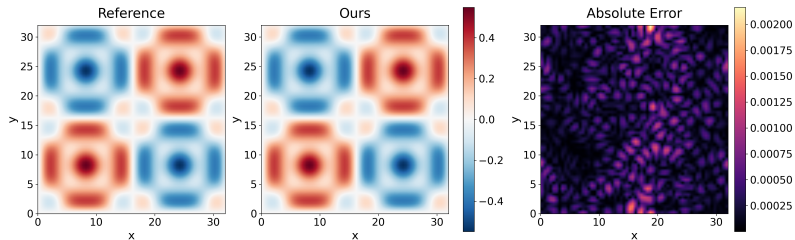


Fig.: Reference solution, CARE-SAV approximation, and pointwise error for the PFC model with $r = -0.5$.

Molecular beam epitaxy: accuracy and energy evolution

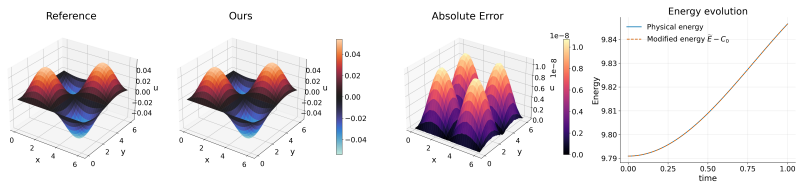


Fig.: Reference solution, CARE-SAV approximation, pointwise absolute error, and energy evolution for the MBE model at $t = T$.

Grain growth: coarsening dynamics

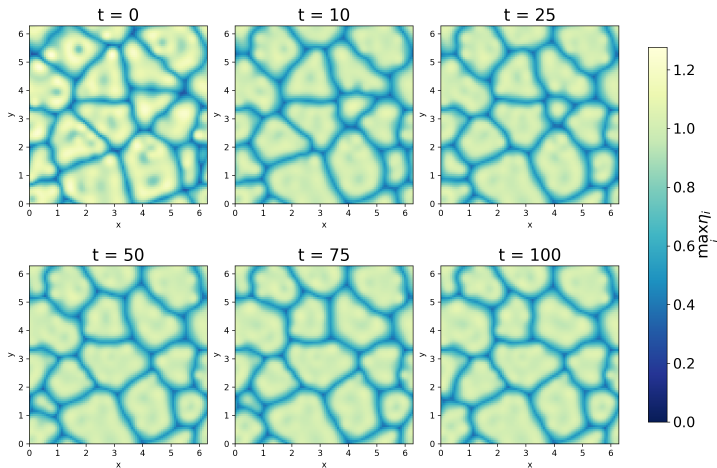


Fig.: Evolution of the maximum order parameter during grain coarsening.

Grain growth: energy and statistical evolution

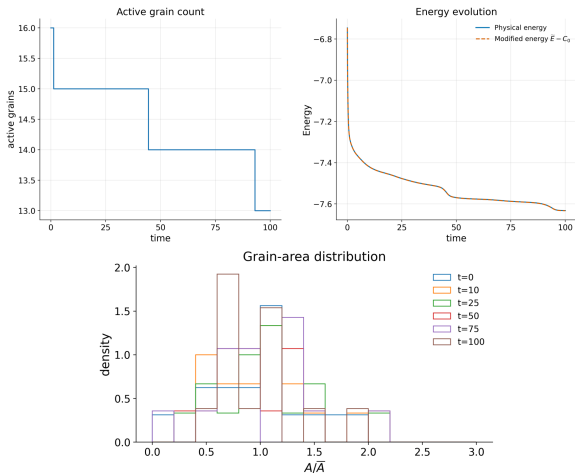


Fig.: Active grain count, physical and modified SAV energies, and normalized grain-area distribution.

Table of Contents

- 1 Motivation
- 2 Gradient flow and SAV
- 3 Conditioning-aware representation enhancement
- 4 The CARE-SAV method
- 5 Main results
- 6 Numerical experiments
- 7 Takeaways**

Takeaways

Main contributions

CARE-SAV enables energy-stable gradient-flow simulation in a compact, conditioning-aware random-feature Galerkin space.

- CARE-SAV combines flexible random features with a conditioning-aware CARE Galerkin space.
- CPQR removes redundant directions and gives an orthonormal reduced basis.
- The fully discrete method is linear, efficient, uniquely solvable, and unconditionally energy stable.
- Numerical results demonstrate accuracy and robustness over diverse gradient-flow models.

Links

Paper Link: <https://arxiv.org/abs/2608.25300>

Personal Homepage: <https://derrose5.github.io>

谢 谢!

Thank you!